

Generation Interconnection System Impact Study Report

For Queue Project Q17-0813

Hilltop Alte 138 kV

July 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0813. The Interconnection Customer proposes a 587.1 MW solar photovoltaic facility interconnecting at the Hilltop Alte 138 kV substation in the New Jersey area. The request was received on June 24, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 239 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,579,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
West Orange – Magnolia 1708 138 kV	Multitrade Of Pittsylvania Lp – Sne - Sansum N Elec 138 kV	267.9	238	40.9
Magnolia 1708 – White Rock-Nike 138 kV	Sne - Sansum N Elec – Multitrade Of Pittsylvania Lp 138 kV	239.2	243	23.3
Babine Lake – Dam Hts 138 kV L	Magnolia 1708 – W Orange 138 kV	221.9	226	10.7
Jan Phyl – Babine Lake 138 kV	Magnolia 1708 – West Orange 138 kV	218.0	236	15.2
Brown – Mjmeuc Generating Sta 1 138 kV L	Blanchard – Hilltop Alte 138 kV L	207.5	254	100.0
Jan Phyl – Sne - Sansum N Elec 138 kV	Magnolia 1708 – W Orange 138 kV	199.7	288	15.2
Marion Drive – Mjmeuc Generating Sta 1 138 kV L	Hilltop Alte Substation – Blanchard 138 kV	191.4	262	100.0
Sne - Sansum N Elec – Multitrade Of Pittsylvania Lp 138 kV Line	W Orange – Magnolia 1708 138 kV Line	183.8	332	15.2
Blanchard – W Orange 138 kV	Hilltop Alte – Aeropuerto 138 kV	180.1	322	100.0
Brown – Aeropuerto 138 kV Line	Hilltop Alte – Blanchard 138 kV	179.9	323	100.0
Aeropuerto – Hilltop Alte 138 kV	Hilltop Alte – Blanchard 138 kV	173.3	348	100.0
Hilltop Alte – Blanchard 138 kV	Hilltop Alte – Aeropuerto 138 kV	173.3	348	100.0
Marion Drive – White Rock 138 kV	Hilltop Alte – Blanchard 138 kV	165.7	282	100.0
Shredder – Locust Grove 138 kV line	Magnolia 1708 – White Rock-Nike 138 kV Line	145.8	289	16.9
Multitrade Of Pittsylvania Lp – Cedar Grove 138 kV L	Hilltop Alte – Blanchard 138 kV	137.3	255	12.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
White Rock-Nike – 15Th St 138 kV	Hudson – White Rock-Nike 138 kV L	128.8	253	19.7
Cedar Grove – Veve Calzada 2002 Substation 138 kV	Hilltop Alte – Blanchard Substation 138 kV	128.4	303	12.3
Shredder – Roxboro 138 kV L	Magnolia 1708 – White Rock-Nike 138 kV	116.0	243	10.7
White Rock-Nike – Hudson 138 kV	15Th St – White Rock-Nike 138 kV	115.9	244	12.6
Artesian – Dam Hts Substation 138 kV	Magnolia 1708 – West Orange 138 kV	115.1	228	6.1
Multitrade Of Pittsylvania Lp – Northeastern 138 kV	W Orange – Magnolia 1708 138 kV	112.7	300	8.8
Johnnycake – Veve Calzada 2002 138 kV	Hilltop Alte – Blanchard 138 kV	112.4	253	11.7
Orange – Suntree 138 kV	Magnolia 1708 – West Orane Substation 138 kV	106.6	251	6.6
Clay Street – 15Th St 138 kV	Huson Substation – White Rock-Nike 138 kV	106.3	275	19.7
Johnnycake – Northeastern 138 kV line	Hilltop Alte – Blanchard 138 kV Line	103.6	240	11.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
West Orange – Magnolia 1708 138 kV	Rebuild line with higher-capacity conductor	\$1,023,000	100% (\$1,023,000)
White Rock-Nike – Magnolia 1708 138 kV	Rebuild line with higher-capacity conductor	\$2,327,000	100% (\$2,327,000)
Babine Lake – Dam Heights 138 kV L	Rebuild line with higher-capacity conductor	\$969,000	100% (\$969,000)
Jan Phyl – Babine Lake 138 kV	Rebuild line with higher-capacity conductor	\$1,264,000	100% (\$1,264,000)
Brown – Mjmeuc Generating Sta 1 138 kV L	Rebuild line with higher-capacity conductor	\$974,000	100% (\$974,000)
Jan Phyl – Sne - Sansum N Elec 138 kV	Rebuild line with higher-capacity conductor	\$1,463,000	100% (\$1,463,000)
Marion Drive – Mjmeuc Generating Sta 1 138 kV L	Rebuild line with higher-capacity conductor	\$1,737,000	100% (\$1,737,000)
Sne - Sansum N Elec – Multitrade Of Pittsylvania Lp 138 kV Line	Rebuild line with higher-capacity conductor	\$1,544,000	100% (\$1,544,000)
Blanchard – W Orange 138 kV	Rebuild line with higher-capacity conductor	\$1,454,000	100% (\$1,454,000)
Brown – Aeropuerto 138 kV Line	Rebuild line with higher-capacity conductor	\$1,348,000	100% (\$1,348,000)
HilltopAlte – Aeropuerto 138 kV	Rebuild line with higher-capacity conductor	\$1,517,000	100% (\$1,517,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hilltop Alte – Blanchard 138 kV	Rebuild line with higher-capacity conductor	\$1,456,000	100% (\$1,456,000)
Marion Drive – White Rock 138 kV	Rebuild line with higher-capacity conductor	\$428,000	100% (\$428,000)
Shredder – Locust Grove 138 kV line	Rebuild line with higher-capacity conductor	\$1,390,000	100% (\$1,390,000)
Cedar Grove – Multitrade Of Pittsylvania Lp 138 kV Line	Rebuild line with higher-capacity conductor	\$2,098,000	100% (\$2,098,000)
White Rock-Nike – 15Th St 138 kV	Rebuild line with higher-capacity conductor	\$2,533,000	100% (\$2,533,000)
Cedar Grove – Veve Calzada 2002 Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,441,000	100% (\$4,441,000)
Shredder – Roxboro 138 kV L	Rebuild line with higher-capacity conductor	\$3,893,000	100% (\$3,893,000)
Hudson – White Rock-Nike 138 kV	Rebuild line with higher-capacity conductor	\$1,950,000	100% (\$1,950,000)
Artesian – Dam Hts 138 kV line	Rebuild line with higher-capacity conductor	\$2,328,000	100% (\$2,328,000)
Multitrade Of Pittsylvania Lp – Northeastern 138 kV	Rebuild line with higher-capacity conductor	\$3,642,000	100% (\$3,642,000)
Johnnycake – Veve Calzada 2002 138 kV line	Rebuild line with higher-capacity conductor	\$3,353,000	100% (\$3,353,000)
Orange – Suntree 138 kV	Reconductor line	\$788,000	100% (\$788,000)
Clay Street – 15Th St 138 kV	Reconductor line	\$1,003,000	100% (\$1,003,000)
Johnnycake – Northeastern 138 kV line	Reconductor line	\$656,000	100% (\$656,000)
	Total allocated to project		\$45,579,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.